Earth-State Embeddings, after the reset:
what a clean holdout leaves of a learned North Atlantic forecaster

blauewelt/earth — technical report, v8 (supersedes v1–v7)

31 August 2026

Abstract

This report replaces seven earlier versions. Their headline results — rolled twelve-month forecasts of the North Atlantic scored at "corridor AUC" 0.93 and above, and every effect measured on them — were produced by models that had been trained on the years they were evaluated on. The training pool admitted a window whenever its *final* target fell outside the held-out years, while the loss was dense over all 144 frames of the window; 21,018 of 400,176 supervised frame-targets were held-out bins. The signatures are unambiguous: field anomaly correlation 0.985 at five days and 0.973 at a year (a forecast decays, a replay does not); skill 0.84 on trained longitudes against 0.18 on held-out ones; an ensemble whose members stay pinned together inside the training record and fan out past it. All of that is withdrawn, and the archived report is kept only as a record.

What survives, measured under a corrected pool that excludes every window touching a held-out frame, is small and specific. Heads from 7.6M to 400M parameters reach their best held-out one-step loss inside roughly the first 2,000 of 200,000 steps and worsen thereafter; the best level sits in 0.58–0.62 of the persistence error across a 53× span in capacity. The first clean twelve-month roll decays like a forecast — corridor anomaly correlation 0.606 at five days, 0.41 at ten, ≈ 0.1 from two months on, −0.03 at a year — and beats raw persistence at every lead but the last. It does *not* beat damped persistence at any lead: the cheapest classical null wins everywhere. Its squared-error score is negative (−0.44 mean MSSS against climatology) because the head emits anomalies at 78% amplitude while its correlation is 10%; an exact identity reproduces all 73 leads to 0.0135, and the same states rescaled would score about +0.02. Surviving correlation lives in sea-surface temperature and height at short lead; 32 of the 40 input channels score nothing at any lead. Transport at 26.5°N shows no measurable skill at roughly nine effective independent starts, and cannot at this sample size. Every clean number is n=1. The report closes with what the data can decide and what it cannot, and the order in which the next experiments should run.

1 Introduction

The programme documented here set out to learn a forward model of the North Atlantic from 43 years of ocean reanalysis: compress the observed state at each grid point into an embedding, predict the embedding forward with a transformer, decode and score. Seven versions of this report described a steady climb in rolled-forecast skill across capacity, spatial coupling, input noise and training-pool changes. On 2026-08-28 a question about where a held-out year's information could be entering the model exposed that the climb was a measurement of memorisation. This version records what happened, what is withdrawn, and the results that were obtained after the protocol was corrected — which are few, and which point the programme somewhere different from where it was going.

It is written to be read without the earlier versions. Section 2 describes the data and Section 3 the models and the evaluation, at the level needed to read the numbers. Section 4 gives the contamination mechanism and the evidence, and Section 5 the corrected protocol. Section 6 is the report's substance: everything measured under the corrected pool. Section 7 sets out the statistical power of the data, which governs what any of this can mean. Section 8 says what should be done next. Appendix A lists the claims of the previous versions and the status of each.

2 Data

The working tensor is `family4_na025_pentad_r2`: the North Atlantic window 0–70°N, 100°W–20°E on a 0.25° grid (281 × 481 cells, 86,698 of them ocean), at five-day (pentad) cadence from 1982-01 to 2024-12

— 3,142 time bins — with 40 channels per cell. Three come from the GLORYS ocean reanalysis (surface current speed, log mixed-layer depth, sea-surface height); 32 are the Roemmich–Gilson Argo temperature and salinity climatology at 16 pressure levels (native 1° and monthly, upsampled, available from 2004); four are NCEP Reanalysis-1 wind stress and its within-bin variability; one is OISST sea-surface temperature. Channels absent before their record begins enter as missing tokens. The truth series for the transport read-out is the RAPID-MOCHA overturning transport at 26.5°N (2004–), deseasonalised.

Two facts about this tensor decide more than any modelling choice. There are 2,417 distinct end-times in the training pool once held-out years and the two-year context window are removed; every one of the 86,698 pixels is a correlated view of the same 2,417 temporal patterns. And the transport target has roughly nine effective independent starts: RAPID is about twenty years long and overturning anomalies decorrelate on months-to-years. These numbers are returned to in Section 7.

The tensor, the truth series and every trained weight cited here are public (Appendix B).

3 Models and evaluation

Stage 1, the codec. A masked autoencoder over the 40 channels of one pixel at one time bin ("PixelMAE": 512-wide, 12 layers, four heads, decoder width 256; 37.976M parameters) produces a 32-dimensional embedding z. It is trained self-supervised on all longitudes with the years 2009, 2017 and 2023 excluded from both its training and its anomaly statistics, then frozen. The codec used for every result here is the checkpoint `run-415__pixelmae.pt`; its embedding of the whole tensor is a $3142 \times 86698 \times 32$ float16 array.

Stage 2, the head. A causal transformer over a window of $K{=}144$ consecutive embeddings at one pixel (two years of pentads) plus a 145-point spatial stencil of neighbouring embeddings on a spiral ring, predicting the embedding one bin ahead. The loss is dense: every one of the 144 window positions is supervised against the embedding one bin after it. Input noise at 0.7 of the persistence scale, gradient clipping at 128, batch 256, learning rate 10^{-3} with exponential decay (half-life 40,000 steps, warm-up 2,000), single-step (non-autoregressive) training. Widths used here: 256×8 (7.598M parameters), 512×12 (40.388M), 1024×16 (206.659M), 1280×20 (399.948M). The held-out one-step loss is reported as the ratio of the head's z-space MSE to the persistence MSE on the same held-out windows (denominator 21.44621 for this tensor and codec); a ratio of 1 is "no better than repeating the last embedding".

The roll. From a true context of 144 bins the head advances every ocean pixel one bin at a time, feeding its own prediction back, for a horizon of 73 steps (365 days). The rolled embeddings are decoded through the frozen codec and scored against the standardised truth anomalies, per pixel and channel, on three nested scopes: **gate** (600 random pixels plus the RAPID section, 864 cells), **corridor** (the fastest quarter of the window by mean current speed, dilated two cells, plus the section: 30,158 pixels), and **window** (all 86,698). Starts are three per held-out year, spread round the seasonal cycle, from the held-out years 2009, 2017 and 2023 — nine starts in all. This "interspersed" holdout is the one every number in this report uses; a terminal holdout is defined in Section 5 and has not yet been spent.

Metrics. For each lead h: $\mathrm{MSSS}_x = 1 - \mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_x$ against three references — **climatology** (x predicts zero anomaly), **raw persistence** (x repeats the last observed anomaly), and **damped persistence** (x decays it with a per-pixel, per-channel AR(1) coefficient fitted on training years); the **anomaly correlation** ACC between rolled and true fields; and the **amplitude ratio** a (rolled std / true std). The quantity earlier versions called "corridor AUC" is the mean $\mathrm{MSSS}_{\mathrm{clim}}$ over the 73 leads. It is not an AUC; a negative value means the squared error exceeds the anomaly variance, not "below chance". The transport read-out regresses the rolled section states onto RAPID (an unpooled, learned read-out) and reports the correlation in three lead bands.

Replication. The stage-2 forecast objective is stable across seeds (pairs at the 206M tier agree to 0.001–0.002 in the one-step ratio), but rolled skill at pentad cadence has never been replicated: every clean rolled number below is a single seed, and is written as such.

4 The contamination

4.1 Mechanism

The training pool built windows of 144 consecutive bins and admitted a window if and only if its final scored bin, the one at position 144, was not in a held-out year. The loss, however, was dense: all 144 positions were supervised. A window ending in January 2010 therefore carried the whole of 2009 in its supervised positions while passing the check. Measured on the real axis, 21,018 of the 400,176 supervised frame-targets in the pool were held-out bins. Every head trained before 2026-08-28 — monthly and pentad, every width, every stencil — had been teacher-forced on tens of transitions per pixel from the years it was then scored on. The authors of the earlier reports (Claude sessions) stated in those reports that the held-out years were held out of training. That statement was false.

The correction is a pool rule with three scopes. `endpoint` is the legacy rule, kept only so the archive reproduces. `target` masks held-out targets from the loss but still admits them as context (−5.25% supervision). `window`, the default from 2026-08-29, admits no window that touches a held-out bin anywhere the forward pass can see it (−13.03% supervision: 209,549,066 training windows against 240,933,742). The pool self-certifies by brute force before training and refuses to start on a violation.

4.2 Four signatures, each from an artefact

1. **Skill that does not decay with lead.** The old 206M head, rolled under the exact battery of Section 3, scores a corridor anomaly correlation of 0.985 at five days, 0.970 at 90, 0.969 at 180 and 0.973 at 365 (Figure 4, left). Its transport correlation *rises* from 0.48 over 5–90 days to 0.58 over 185–365. No forecast does this; a replayed trajectory does.

2. **No spatial generalisation.** On the tensor variant with a longitude block (45–25°W, 80 columns) held out of training, the same kind of head scores mean $\mathrm{MSSS}_{\mathrm{clim}}$ 0.838 on trained corridor pixels and 0.176 on held-out ones (gate scope: 0.821 against 0.330; the earlier stencil-free gate head: 0.804 against 0.058). On ocean it has not seen, the head removes 6–18% of the anomaly variance. Every blended headline in the earlier versions mixed roughly a quarter of untrained pixels into the rest.

3. **Confidence exactly where it memorised.** An eight-member stochastic ensemble from the same head, rolled 36 months from a context ending 2004-12 — inside the training record — keeps a member spread (standard deviation of the decoded field, corridor mean) of 0.099 over the first year and 0.137 over the third. Rolled 36 months past the end of the record it spreads to 0.360 and 0.532 (Figure 1). No truth enters this measurement.

4. **The pool arithmetic itself**, above.

4.3 What is withdrawn

Every rolled number produced by a head trained under the endpoint rule, as evidence of forecast skill: the monthly champion line and its seed pairs, the capacity, noise and longitude-holdout effects measured on rolled scores, the pentad and daily rolls, the long hindcasts, and the ensemble results. Retained, because they do not depend on the stage-2 pool: the data description, the codec's architecture and reconstruction behaviour, the seed-replication arithmetic on the one-step objective, and the JAX reimplementation's parity against the PyTorch code. Appendix A lists the earlier claims individually.

5 The corrected protocol

Fixed on 2026-08-30 and applied to everything that follows:

- Window-scope pool, self-certified.
- Lead-decay as a standing falsifier: a rolled profile that does not decay with lead is a replay, whatever its level.

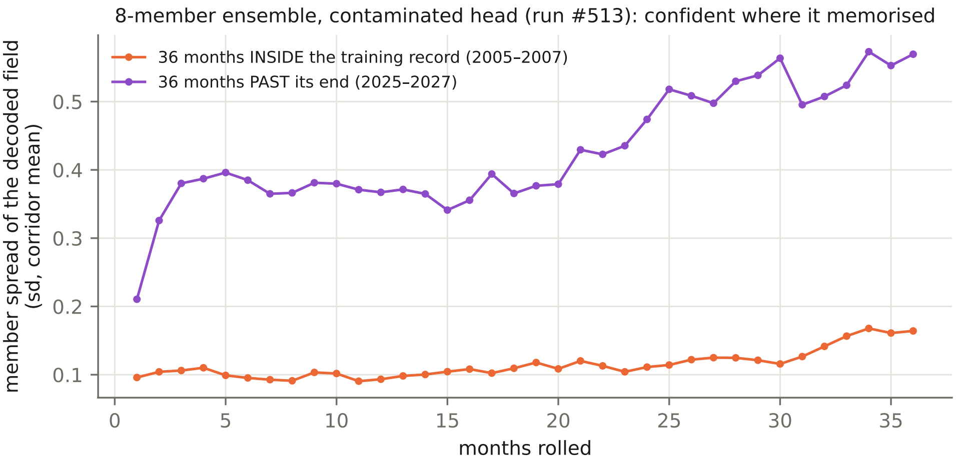

Figure 1: Member spread of an eight-member ensemble rolled from a head trained under the contaminated pool (run #513), 36 months inside the training record and 36 months past its end. The dispersion is a property of the rolled states alone; a model that had learned dynamics would spread on both.

- Trained- and held-out-longitude scores reported separately, never blended.
- A classical null beside every number, with the null named.
- Early stopping at the held-out minimum; the small tier (7.6M) as the default, larger widths needing a written argument.
- Rolled skill is the verdict; a probe of the current state is not.
- A *terminal* holdout for the final test — train $\leq$ 2020, test 2021–2024, no gap — to be opened once. Because the held-out years are a property of the codec (the stage-2 pool inherits them from the codec's configuration), the terminal test requires a codec trained on $\leq$ 2020; none exists yet, and nothing in this report touches 2021–2024 as a test set.

6 Results under the corrected pool

6.1 Training dynamics: the held-out minimum is early, at every width

E-059 is the old 206.66M head retrained with one change, the pool. Its training loss tracks the contaminated twin's closely (both reach a ratio of 0.02–0.03 at 200k steps), while its held-out ratio does not fall at all after the first record: 0.6105 at step 2,000, 0.6411 at 20,000, 0.6859 at 200,000, against the contaminated twin's monotone descent 0.2247 $\rightarrow$ 0.0812 $\rightarrow$ 0.0330 on identical validation windows (Figure 2, left). The train/held-out gap of the clean head at the end of training is a factor of thirty.

The width ladder (E-060) repeats the corrected run at 7.598M, 40.388M and 399.948M parameters with every other knob identical and a learning-rate schedule that traces the same trajectory step for step. All four arms reach their best held-out ratio inside 2,000 steps and get worse afterwards (Table 1, Figure 2, right). The best levels span 0.578–0.621 across a 53$\times$ span in parameters. At the pre-registered comparison step of 20,000, the 7.598M arm reads 0.692 against the 206.659M arm's 0.641: the prediction, made before the arms ran, that the small arm would land within 0.02 of the large one failed by 0.051 and is recorded as failed. Width buys how fast an arm reaches its minimum and how badly it then degrades, not the level of the minimum.

The in-training transport probe (a pooled ridge read-out of the last position's hidden state against RAPID, computed every 2,000 steps) tells the capacity story from the other side: the 7.598M arm holds 0.598–0.611 across ten probes, the 40.388M arm drifts to 0.589, and the 206.659M arm falls from 0.616 at 20k to 0.522 at 200k (Figure 3). This is a probe and not a verdict, but the direction is the one a memorising model would show.

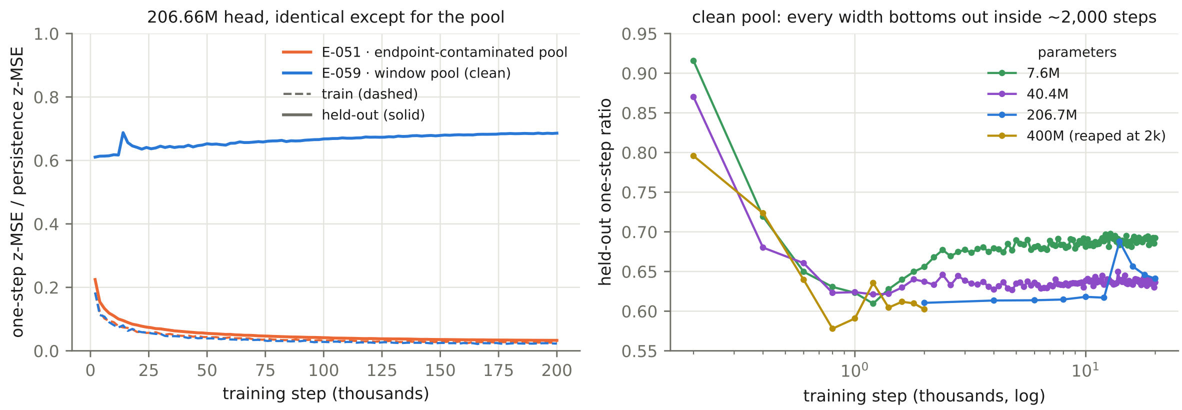

Figure 2: Left: one-step z-space MSE relative to persistence, training (dashed) and held-out (solid), for the 206.66M head under the contaminated pool (E-051) and the corrected pool (E-059) — same architecture, seed, codec, embedding, validation windows and hardware. Right: the held-out ratio over the first 20,000 steps for four widths under the corrected pool.

head	params	best held-out ratio	at step	ratio at 20k	train ratio at end
E-060a	7.598M	**0.6095**	1,200	0.6922	0.146 (20k)
E-060b	40.388M	0.6212	1,200	0.6363	0.086 (20k)
E-059	206.659M	0.6105	$\leq$ 2,000	0.6411	0.023 (200k)
E-060c	399.948M	0.5779	800	— (stopped at 2k)	0.171 (2k)
E-051 (contaminated)	206.659M	0.0330	200,000	0.0812	0.026 (200k)

Table 1: Held-out one-step ratio under the corrected pool by width. E-059's first record is at step 2,000, so its minimum is at or before it; the smaller arms, recorded every 200 steps, bottom out at 1,200.

6.2 The first clean roll

The E-059 head at its 200k end state was rolled through the battery of Section 3 (run #516), protocol-identical to the roll of its contaminated twin (run #510). Table 2 gives the corridor numbers by lead; Figure 4 shows the two heads side by side and the clean head against its three references.

Four things are established by this roll, and they are the report's main results.

It decays like a forecast. Anomaly correlation 0.606 at five days, 0.412 at ten, 0.265 at fifteen, 0.20 at a month, near 0.1 from two months on, -0.03 at a year — against the contaminated twin's flat 0.97–0.99. The lead-decay falsifier of Section 5 passes. The corrected pool removed the replay.

It beats raw persistence, and loses to damped persistence at every lead. $\mathrm{MSSS}_{\mathrm{pers}}$ is positive at 72 of 73 leads (mean $+0.204$). $\mathrm{MSSS}_{\mathrm{damped}}$ is negative at all 73, from -0.011 at five days to -0.719 at a year (mean -0.461; gate -0.416, window -0.420). A per-pixel AR(1) decay toward climatology, fitted on training years and costing nothing, is a better forecast of this field than the transformer at every horizon. This is the fact the next experiments are designed around.

The negative squared-error score is amplitude, not sign. For a forecast with anomaly correlation ACC and amplitude ratio a against a zero-anomaly reference, $\mathrm{MSSS}_{\mathrm{clim}} = 1 - (1 + a^2 - 2a\,\mathrm{ACC})$ identically. Evaluated from the head's own per-lead ACC and a, this reproduces the measured 73 corridor values to a mean absolute error of 0.0135 (Figure 5). The head emits anomalies at $a \approx 0.78$ long after its correlation has fallen to 0.1; the squared error of a confidently wrong sign is what the negative score measures. The same rolled states rescaled to $a = \mathrm{ACC}$ would score ACC^2 at each lead, a corridor mean of about $+0.02$. That is small, but it is positive, and it is a decoding change rather than a capacity one. Whether a calibration fitted on training-year starts transfers to held-out starts has not been tested.

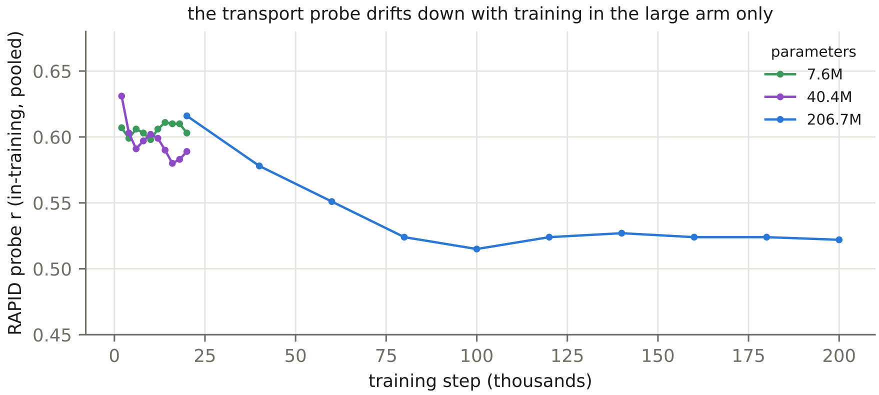

Figure 3: The in-training RAPID probe against training step, by width, under the corrected pool.

lead (days)	$\mathrm{MSSS_{clim}}$	$\mathrm{MSSS_{pers}}$	$\mathrm{MSSS_{damped}}$	ACC	a
5	+0.365	+0.194	−0.011	0.606	0.692
10	+0.109	+0.252	−0.131	0.412	0.673
15	−0.172	+0.178	−0.354	0.265	0.761
30	−0.168	+0.252	−0.244	0.204	0.672
60	−0.320	+0.201	−0.342	0.107	0.690
90	−0.322	+0.211	−0.333	0.132	0.725
180	−0.354	+0.337	−0.357	0.153	0.778
270	−0.479	+0.227	−0.479	0.009	0.720
365	−0.719	−0.101	−0.719	−0.031	0.822
mean of 73 leads	−0.439	+0.204	−0.461	0.105	0.780

Table 2: The clean 206.66M head (E-059) rolled from nine starts in 2009, 2017 and 2023, corridor scope. Gate and window scopes read mean $\mathrm{MSSS_{clim}}$ −0.395 and −0.401, mean ACC 0.131 and 0.121. Single seed.

Where correlation survives is specific. Only eight of the forty channels are scoreable; the 32 Argo channels carry no scored samples at any lead. Of the eight, sea-surface temperature holds correlation longest (0.76 at five days, 0.34 at 30 and at 90) and is the only channel whose $\mathrm{MSSS_{clim}}$ stays positive beyond a few pentads; sea-surface height is the sharpest at one step (0.84) and falls below 0.2 by a month; current speed and mixed-layer depth are one-step quantities; the four wind channels are essentially noise past one step (Figure 6, Table 3).

6.3 Transport at 26.5°N

The unpooled read-out of the rolled section states against RAPID gives $r = +0.107$ over 5–90 days (n=162), −0.242 over 95–180 (n=129), +0.163 over 185–365 (n=150). At nine effective starts these are inside the noise of zero and are read as “no measurable transport skill”, never as a sign. The contaminated twin’s 0.48/0.57/0.58 on the same starts is the recall curve of Section 4. A separate 36-month free run from a context ending 2004-12 — entirely inside the training years, so a memorisation-free reading is not available from it — correlates 0.636 with the true transport at pentad resolution and −0.13 on an 18-month low-pass, at 61% amplitude: whatever the head tracks in the training years lives at high frequency, and the multi-year band that is the overturning question is not measured by this roll. The head is deterministic, so no dispersion test of the kind in Figure 1 can be made for it; that test is unanswered for clean heads, not failed.

6.4 What the second clean head will add

The 7.598M head (E-060a) has been mirrored to the public release and is being rolled through the identical battery as this report is written. It will give the width axis at two points under a clean pool on the rolled field — the question Table 1 answers only for the one-step loss. Its reading was registered

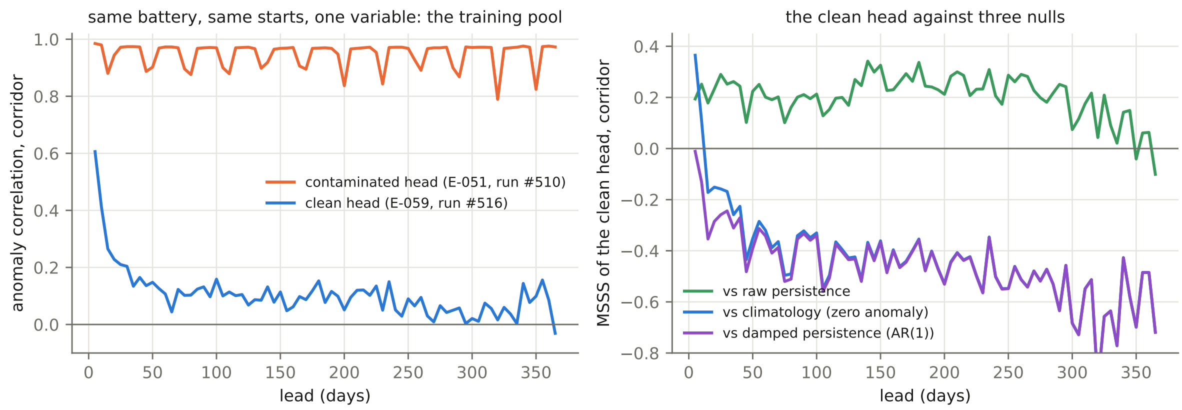

Figure 4: Left: corridor anomaly correlation against lead for the contaminated head (run #510) and the clean head (run #516), same battery, same nine starts. Right: the clean head's MSSS against climatology, raw persistence and damped persistence.

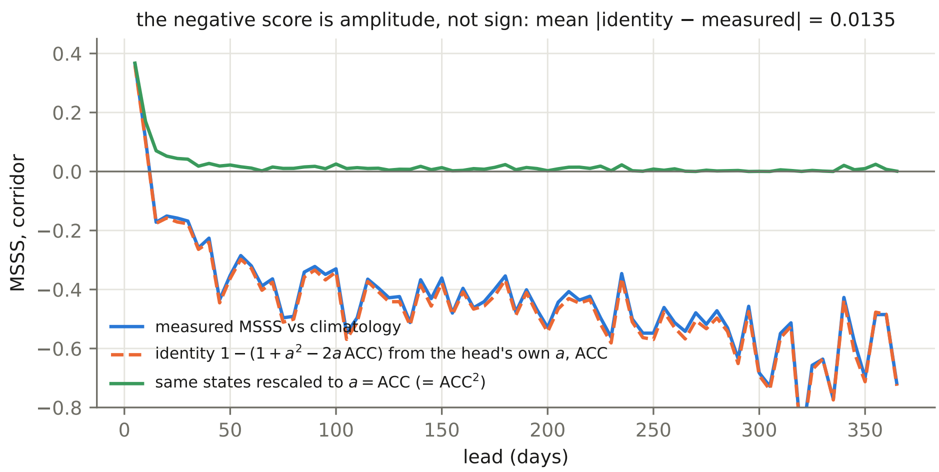

Figure 5: Measured corridor $\mathrm{MSSS}_{\mathrm{clim}}$ per lead for the clean head, the value the identity predicts from the head's own ACC and amplitude ratio, and the value the same states would score rescaled to $a = \mathrm{ACC}$.

before it ran: the lead-decay falsifier must pass; the level is read against Table 2; parity or better would mean capacity was never the axis on the rolled field either.

7 Statistical power

Three numbers govern how much any result above can mean.

2,417 end-times. The pool's 209,549,066 training windows are $2{,}417 \times 86{,}698$. Consecutive windows share 143 of 144 frames. With 206M parameters there are roughly 85,000 parameters per distinct temporal pattern; the corrected pool removed the reward for memorising the held-out years, not the incentive to memorise the training ones. Early stopping, small widths and regularisation are palliative. Only new temporal samples change this number; finer grids give more correlated views of the same 2,417 patterns and are declined.

Nine effective starts. The transport bands above have n of 129–162 pentads from nine starts; the independent-sample count for a target that decorrelates over months-to-years is of order ten. A difference of 0.2 in a band correlation is not resolvable. No architecture, corpus or resolution changes this; it is a

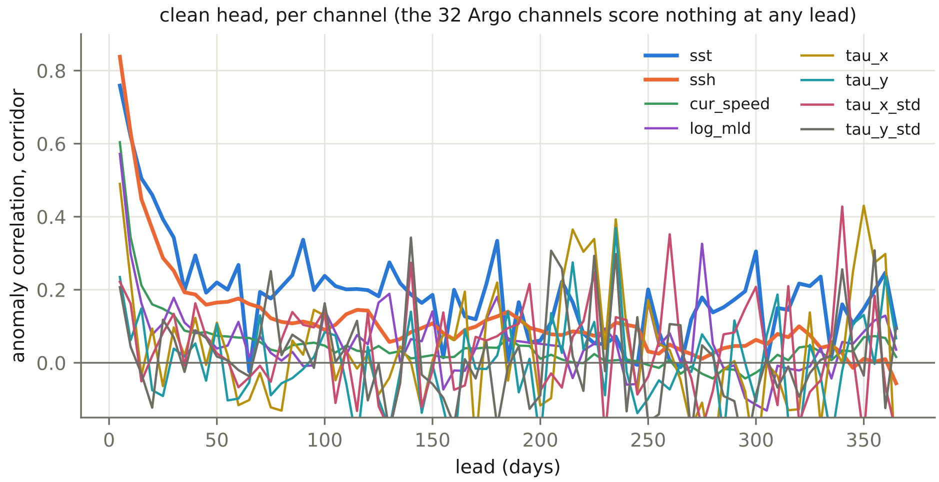

Figure 6: Corridor anomaly correlation against lead, per scoreable channel, for the clean head.

channel	ACC 5 d	ACC 30 d	ACC 90 d	mean $\mathrm{MSSS}_{\mathrm{clim}}$	mean $\mathrm{MSSS}_{\mathrm{damped}}$
`sst`	0.759	0.343	0.337	−0.165	−0.191
`ssh`	0.838	0.252	0.113	−0.499	−0.562
`cur_speed`	0.604	0.131	0.052	−0.591	−0.606
`log_mld`	0.572	0.178	−0.009	−0.464	−0.469
`tau_x`	0.490	0.097	0.022	−0.383	−0.386
`tau_y`	0.235	0.039	−0.017	−0.426	−0.428
`tau_x_std`	0.222	0.134	0.104	−0.409	−0.410
`tau_y_std`	0.207	0.070	0.057	−0.465	−0.466

Table 3: Per-channel corridor scores for the clean head. The 32 `rg_*` channels are omitted: no scored samples at any lead.

property of RAPID's length and the overturning's timescale.

n=1. Every rolled number in this report is a single seed at a cadence where rolled skill has never been replicated. The one-step objective is stable across seeds; the rolled scores may not be. No two clean rolled numbers can yet be called different, including the two heads of the width axis.

Together these say that the field-level questions — where does a learned model beat classical nulls, and out to what lead — are answerable with this data, with intervals, and the transport question at 26.5°N is not. The report's recommendation is to make the former the headline and to report the latter as a bound.

8 What should happen next

In order, and with nothing trained before the first two exist.

1. **The null ladder on the identical battery.** Damped persistence as a row of its own; nearest-analogue retrieval from training-year states (if it matches the transformer, the transformer is a lookup table); a Linear Inverse Model on the leading EOFs of the training-year anomaly fields, cross-validated inside the training years, and a second LIM in the codec's embedding space — the difference between the two is the first measurement of whether the codec discards predictable signal. The pre-registered expectation is that the LIM matches or beats both clean heads at every lead beyond the first; if it does, that is the programme's central result.
2. **Intervals.** Block-bootstrap confidence intervals over (year, start) blocks on every per-lead score and every difference; a rolling-origin blocked cross-validation over several held-out years for de-

velopment, keeping 2021–2024 untouched; a minimum-detectable-effect table so that “cannot be resolved” becomes a stated result.

3. **Short budgets and replicates.** Stage-2 training capped near 5,000 steps with checkpoints every 500 and selection at the held-out minimum, which makes a 7.6M arm a ∼1.5-hour job and five seeds the default.

4. **The terminal codec**, trained on ≤ 2020, with its embedding of the tensor; the long pole, started early.

5. **Then, cheaply:** roll an early-stopped checkpoint against the end state; five seeds; amplitude calibration fitted on training-year starts and applied to held-out ones, for nulls and heads alike; the head trained on the eight scoreable channels only.

6. **Data** that touches the bound and nothing that does not: the Argo channels to a coarse sidecar; the record extended forward to 2026 and, as a measurement, backward toward 1958 at 1°; surface flux forcing as missing physics; climate-model output, if at all, only to pretrain the codec and with a masked-reanalysis control.

A Claims of the earlier versions and their status

claim (v1–v7)	status
Rolled twelve-month corridor “AUC” 0.93–0.94 for the best monthly heads, with seed pairs agreeing to 0.0003–0.005	Withdrawn: contaminated pool. The seed agreement was real and measured the replay, not skill.
Capacity +0.042, input noise +0.045/+0.050, all-longitude training +0.071 on the rolled score	Withdrawn: effects measured on a contaminated metric. Capacity re-measured clean: no effect on the held-out minimum (Table 1).
Pentad one-step ratio 0.0330 at 200k as “the best pentad number on record”	Withdrawn as evidence of forecast skill: the one-step gain was largely the held-out-year memorisation term (Figure 2).
Long hindcasts correlating with RAPID over years	Withdrawn: calendar replay, as v7 itself half-acknowledged.
“Corridor AUC” as a metric name	Retired: the quantity is mean MSSS against climatology.
Quantised (FSQ) codec as the best forecasting substrate (0.4394 vs 0.5056)	Withdrawn as a skill claim: both sides contaminated. The comparison of substrates may be re-run under the corrected pool.
Probe results on the frozen codec (attention head r=0.672; raw-pixel control 0.659)	Not affected by the stage-2 pool; not reproduced here because a probe of the current state is not a forecast verdict under the corrected protocol.
The JAX reimplementation’s parity gates against PyTorch	Stands.
The data description and the codec architecture	Stand; summarised in Sections 2–3.

The archived report is at `ml/paper/archive/v7-2026-08-27-contaminated/` with a note explaining its retirement.

B Data and code availability

All code is at `https://github.com/blauewelt/earth`; the evaluator is `ml/rollout_spatial.py`, the stage-2 trainer `ml/temporal.py` and its JAX port `ml/jaxport/`. Result bundles for every run named here are on the repository’s `ml-metrics` branch as `probes-<n>.json` (#510, #513, #516); trained weights are on the `model-checkpoints-v1` release (`run-415__pixelmae.pt`, `head-weights-e059-200k-window-s0.pt`, `head-weights-e060a-20k-window-s0.pt`, `head-weights-e060b-20k-window-s0.pt`); the tensor is on the `data-cache-v1` release and the Hugging Face dataset `chfrank/earth-tensors`; the codec’s embedding of it is on the `embed-cache-v1` release. The figures in this report are generated by `ml/paper/make_figs.py` from `ml/paper/data/reset_data.json`, which `ml/paper/extract_data.py` builds from those public bundles and the training records; no number in a figure or table is typed in.

Sources: GLORYS12 and the GREP ensemble (E.U. Copernicus Marine Service Information; use is subject to the CMEMS licence and requires crediting the service); the Roemmich–Gilson Argo climatology (Scripps Institution of Oceanography; Argo data are collected and made freely available by the International Argo Program and the national programs that contribute to it); NCEP/NCAR Reanalysis-1 (NOAA PSL); OISST v2.1 (NOAA NCEI); the RAPID-MOCHA-WBTS transport time series (funded by NERC, NSF and NOAA). The CMIP6 pre-industrial-control corpus mentioned in Section 8 is built from HadGEM3-GC31-MM (Met Office Hadley Centre) and CNRM-CM6-1-HR (CNRM / CERFACS) output via the public CMIP6 archive on Google Cloud; it is not used in any result here.